

Supplementary material for “Discourse about welfare benefits”

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This document provides supplementary details for the paper. Section, figure and table numbers without an S prefix refer to the paper and its appendix.

S1 Welfare trope definitions

Table S1 reports the full definition of each of the fourteen welfare tropes described in appendix B, including the boundary notes supplied to the GPT classifier.

Table S1: Full definitions of the fourteen welfare tropes, including the boundary notes supplied to the GPT classifier.

Trope	Definition
The Fake Disability Claimant	People who claim disability, incapacity, or mobility benefits for supposed physical limitations while not actually being disabled. The core narrative is the contrast between claimed incapacity – inability to walk, severe back pain, being wheelchair-bound – and evidence that the claimant is not genuinely disabled. This might include surveillance footage or social media showing strenuous physical activities such as playing sports, running marathons, dancing on stage, or skydiving, but could also involve other evidence exposing the false claim. <i>Boundary note:</i> Distinct from the Secret Worker cluster, which is about concealing employment income. Where the concealed work also contradicts a disability claim (e.g., driving a lorry while claiming immobility), assign c5 if paid employment is the primary deception revealed; assign c1 if the text emphasizes recreational or leisure activity contradicting the medical claim.

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Trove	Definition
The Secret Worker	Individuals who claim unemployment, incapacity, or other out-of-work benefits while secretly holding paid employment or running a business. The fraud is specifically about concealing earned income – working as cleaners, delivery drivers, exotic dancers, builders, online sellers, or other occupations while simultaneously collecting benefits intended for those unable to work. Unlike the Fake Disability Claimant (caught doing leisure activities that contradict medical claims), these individuals are caught holding actual jobs. Unlike the Hidden Wealth Fraudster (who conceals passive assets), these individuals are actively earning undeclared income. <i>Boundary note:</i> Where the concealed work also contradicts a disability claim, assign whichever cluster captures the primary focus of the narrative — c5 if the story is mainly about the hidden employment, c1 if it is mainly about the false disability claim.
The Wealth ster	Hidden Fraud- Individuals who fraudulently claim means-tested benefits while secretly possessing substantial financial assets – hidden savings accounts, undisclosed property, inheritances, lottery or gambling winnings, insurance payouts, or other windfall income. The fraud is specifically about concealing one’s true financial position from the benefits system rather than faking a medical condition or hiding employment. Stories feature the revelation of assets that were deliberately hidden or not declared. Includes cases where lottery winners fraudulently continue to claim benefits by failing to declare life-changing sums. <i>Boundary note:</i> This cluster requires fraud through active concealment — the claimant must be hiding or failing to declare their true financial position. If wealth or a windfall is publicly known and not being concealed, the article does not belong in c2. An individual who is simply wealthy while receiving benefits, without fraudulent concealment, may fit c3 (lavish lifestyle) instead.

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Trope	Definition
The Concealed Partner Fraud	Individuals who falsely claim to be single, separated, or living alone in order to receive higher benefit payments – typically lone-parent premiums, housing benefits for single occupants, or income support at single-person rates – while actually cohabiting with a partner or spouse. The fraud is exposed through social media (wedding photos, relationship posts), anonymous tip-offs from neighbors, or official records that contradict the claimed living arrangements. The trope centers on the manipulation of household and relationship status for financial gain. <i>Boundary note:</i> Distinct from identity fraud (which involves fabricated personas or stolen identities) and from general financial fraud (which involves hiding assets). The defining feature is the concealment of a domestic partnership.
The Identity Fraud Ring	Organized or large-scale benefit fraud involving fabricated identities, forged documents, stolen personal data, coordinated fraud rings, or claims made on behalf of fictitious people. This includes inventing non-existent children or dependents to claim benefits, using deceased persons’ identities, and creating entirely fabricated personas. Unlike individual claimants who misrepresent their own circumstances, these stories emphasize the systematic, criminal-enterprise nature of the fraud – multiple false claims, networks of accomplices, forged paperwork, and sometimes cross-border operations. The trope frames benefit fraud as organized crime rather than individual moral failing. <i>Boundary note:</i> Distinct from c7 (which involves fabricating illness in real children, not inventing fictional ones). Distinct from c6 (individual misrepresentation of household status). Distinct from c14 (which specifically involves government employees).
The Prolific Parent	Parents – usually mothers, sometimes fathers – with very large families who live entirely on benefits, are perceived as having children irresponsibly at taxpayer expense, and often demand larger state-funded housing to accommodate their growing broods. The trope frames children as a revenue stream rather than a responsibility, emphasizing the financial burden on taxpayers and the perceived moral failing of continued reproduction without financial independence. Stories typically cite the exact number of children, itemize the annual cost to the state, and include commentary about sterilization, benefit caps, or reproductive irresponsibility. <i>Boundary note:</i> Distinct from general benefit lifestyle stories because the number of children is the central narrative element, not spending habits or fraud.

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Trope	Definition
The Criminal on Benefits	Benefit recipients who are involved in serious criminal activity beyond benefit fraud itself, including but not limited to drug dealing, terrorism, violent crime, sex offenses, money laundering, or extremism, where the criminal conduct is the central point of the article and benefits are framed as funding or enabling the criminal behavior. The trope links welfare dependency to broader criminality, suggesting that benefits subsidize dangerous individuals. The threshold test: the criminal conduct must be the narrative focus, not merely incidental background about a person who happens to receive benefits. The criminal activity must be something beyond benefit fraud itself — if the only crime described is fraudulently claiming benefits, the article does not belong in c10.
The Toothless System	Texts where the welfare system’s inability to impose meaningful consequences for fraud is the central narrative – lenient sentencing of fraudsters, absurdly long repayment schedules (repaying at trivially small weekly amounts over decades), bureaucratic errors that allowed fraud to continue unchecked, and institutional failures in oversight or enforcement. The outrage is directed at judges, policymakers, sentencing guidelines, or institutional dysfunction rather than at specific recipients. The defining feature is that the system’s failure is the subject of the text, not a backdrop to an individual fraud case. <i>Boundary note:</i> Distinct from c12, which focuses on individual attitudes and moral judgments about specific people. If a text names a specific individual as the primary subject and includes policy criticism incidentally, assign to the cluster matching the individual’s trope — or c0 if the individual’s case does not match any trope. A lenient sentence mentioned at the end of a fraud story does not make it c13. Assign c13 only when the systemic failure is the story.

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Trove	Definition
The Hardworking Taxpayer vs. The Scrounger	Texts that construct a moral dichotomy between deserving workers who contribute to the system and undeserving claimants who extract from it. The text must contain an explicit contrast between a hardworking or deserving figure and the benefit recipient — moral condemnation of a claimant alone, without a contrasting comparator (e.g. an injured officer, a working family, a taxpayer), is not sufficient for this cluster. This includes stories of hardworking individuals or families who refuse benefits as implicit moral exemplars contrasted against benefit recipients, opinion pieces expressing taxpayer resentment that explicitly juxtapose workers and claimants, and personal testimonies from struggling workers compared against those on welfare. <i>Boundary note:</i> If the text names or describes a specific individual or family and frames them through the deserving/undeserving lens, assign c12 even if policy implications are mentioned. If the text is primarily about abstract policy, sentencing norms, or institutional design with no named individual as the subject, assign c13.
The Concealed Death	A distinctly macabre trope in which individuals hide the death of a relative — parent, spouse, sibling, or housemate — in order to continue collecting their pension, disability, or housing benefits. Stories feature gruesome details of body concealment: corpses kept in bedrooms, buried in gardens, stored in freezers, or hidden under rubble, sometimes for years. The narrative combines benefit fraud with horror, emphasizing both the financial greed and the grotesque disrespect shown to the deceased. <i>Boundary note:</i> Distinct from fabricated dependents (which involves living children). The defining feature is the concealment of a death and the continued collection of a deceased person's benefits.
The Lavish Lifestyle on Benefits	Benefit recipients portrayed as living extravagant, comfortable, or conspicuously consumptive lifestyles funded by the welfare state — occupying expensive homes, taking luxury holidays abroad, driving flashy cars, spending on cosmetic surgery, designer goods, or leisure activities far beyond necessity. The narrative tension comes from the perceived injustice of taxpayers funding a lifestyle they themselves cannot afford. The lavish lifestyle must be portrayed as a direct consequence of benefit generosity — the implicit argument is that benefits are too generous. If the lifestyle is funded by proceeds of crime (including benefit fraud) rather than by legal benefit payments, the article does not belong in c3. The article may mention other conduct incidentally, but the lavish living itself must be attributed to the benefits system.

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Trope	Definition
The Immigrant Welfare Exploiter	Narratives that frame benefit receipt specifically through an immigration lens – immigrants, asylum seekers, refugees, or foreign nationals claiming benefits, organizing migration for welfare access, or being housed in expensive accommodations at taxpayer expense. The trope combines anti-welfare sentiment with anxieties about national identity and belonging, often contrasting the ease with which newcomers obtain support against the struggles of long-standing citizens and taxpayers. Some stories involve fraud; others involve legal entitlements presented as morally unacceptable. <i>Boundary note:</i> Distinct from identity fraud rings (even if immigrants are involved, this cluster focuses on the immigration framing). Distinct from the Criminal on Benefits cluster (unless the narrative centers on immigration status rather than criminal conduct). The immigrant must be the person claiming benefits — if an immigrant is present in the story but is not themselves a benefit claimant, the trope does not apply.
The Fabricated Illness in Children	Parents or carers who invent, exaggerate, or induce illnesses or disabilities in their real children in order to claim disability, carer’s, or other benefits. This includes fabricating cancer diagnoses (shaving children’s heads, faking medical records), claiming children have severe disabilities they do not have, or subjecting children to unnecessary medical treatments. The trope often has Munchausen-by-proxy overtones and is particularly emotive because it involves direct harm to or exploitation of real children as instruments of fraud. <i>Boundary note:</i> Distinct from identity fraud (c9), which covers the invention of entirely fictional or non-existent children to claim benefits – that is a paperwork fraud, not child exploitation. The defining feature of c7 is the abuse or exploitation of real children through medical fabrication.
The System Insider Abuser	Government employees, benefits administrators, fraud investigators, or other system insiders who exploit their positions of trust to commit benefit fraud. This includes eligibility specialists approving false claims for accomplices, fraud investigators themselves claiming disability, public employees fabricating records, and officials using insider knowledge to game the system. The trope highlights institutional corruption and the irony of those appointed to safeguard public funds being the ones who steal them. <i>Boundary note:</i> Distinct from identity fraud rings (which operate from outside the system) and from general fraud (where the claimant has no privileged access). The defining feature is the abuse of institutional position and trust.

S2 Prompts

Table S2 gives the full text of every prompt used in the natural language processing pipeline described in appendix B. Text in braces, such as {name}, is filled in for each article.

Table S2: Prompts used in the natural language processing pipeline.

Classifier	Model	Prompt
<i>Training data</i>		
Translation of UK labelled data (France, Germany)	GPT-4o-mini	"Translate the following string, where fields are separated by '*****', from English into {lang} and return a JSON object in this format:\n" "{\n" " headline: \"translated headline\", \"\n" " article: \"translated article\", \"\n" "}\n" "The order of fields is: headline, article."
<i>Frame</i>		
Individual: about one individual or family	GPT-4o-mini	Respond 'yes' or 'no': is this news article mainly about one specific individual or family? If the article is about multiple individuals or families, respond 'no'. Only say 'yes' if this individual or family is the main focus of the article. If more than half of the article is not about this individual or family, respond 'no'.
Individual: politician or famous	o4-mini	Respond 'yes' or 'no': Identify the main individual/family mentioned in this article. Respond 'yes' if they are a politician or they are famous. Respond 'no' otherwise. If they are only known because they have accused of/committed a crime, then respond 'no'.
Policy	GPT-4o-mini	Respond `yes' or 'no': is this news article mainly about details of a public policy, the implementation of a policy, or a proposed new policy? Only say 'yes' if it is the main focus of the article. If more than half of the article is not about a policy, respond 'no'.

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Classifier	Model	Prompt
Statistics: data, statistics or market brief	GPT-4o-mini	Respond 'yes' or 'no': is this news article mainly about data, statistics or a market brief? Only say 'yes' if it is the main focus of the article. If more than half of the article is not about data, statistics or a market brief, respond 'no'. If the article is mainly about the cost of a public policy, respond 'no'.
Statistics: research report	GPT-4o-mini	Respond 'yes' or 'no': is this news article mainly about a research report? Only say 'yes' if it is the main focus of the article. If more than half of the article is not about a research report, respond 'no'.
<i>Slant</i>		
Slant	o4-mini	Respond 'good', 'bad' or 'neutral': what impression does this article predominantly give about people seeking or claiming welfare benefits? Only focus on the attitude towards people seeking/receiving benefits, not any other sentiment. Answer 'neutral' if it is not clear
<i>Person characteristics</i>		
Main individual	GPT-4o-mini	Identify the main individual, couple, or family in this article. Return only their name. In general, the person who is writing the article is NOT the main individual - instead the main individual is the person who is the subject of the article. If one member of the couple or family is mentioned more than others, return only that individual. If their name is not given, return a short description of the individual eg. "the sister".
Gender	GPT-4o-mini	Return only "man", "woman" or "unclear": According to this article, what is {name}'s gender? Return "man" if they are referred to with male pronouns, and "woman" if they are referred to with female pronouns.
Employment status	o4-mini	Return only "employed", "not employed", "retired" or "unclear": According to this article, what is the work status of {name}? Say "unclear" if it is not mentioned. Do not guess. Receiving benefits does not imply they are not employed.

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Classifier	Model	Prompt
Marital status	GPT-4o-mini	Return only "married", "single", "unmarried partner", or "unclear": According to this article, what is the marital status of {name}? Say "unclear" if it is not mentioned. Do not guess. If they are described as being as husband or wife, or having a husband or wife, say "married". Otherwise, if they are described as having a partner, or being in a couple, say "unmarried partner". If they are described as a single parent, say "single".
Has children	o4-mini	Return only "yes", "no" or "unclear": According to this article, does {name} have children? Say "yes" if {name} is referred to as a mum, dad, parent or grandparent. Say "yes" if {name}'s sons, daughters, children, or grandchildren are mentioned. Only say "no" if the article explicitly says {name} has no children. Return "unclear" if the article contains no information about whether {name} has children. Do not guess.
Number of children	GPT-4o-mini	Return only "number" or "unknown": How many children does {name} have? Return the number as an integer. Include unborn children in the total. Return "unknown" if not mentioned, or a total number is not given. Do not guess.
Native to the country	o4-mini	Return only "yes", "no" or "unclear": Is {name} native to {country}? Return "yes" or "no". Return "yes" if unclear.
Benefit fraud	GPT-4o-mini	Return only "yes" or "no": Does this article concern a case of welfare benefit fraud? Only say yes if someone is, or has been accused of, claiming illegally. If the article is only saying the person is undeserving of benefits, say "no". If the article is about another illegal activity, say "no".
Years claiming	o4-mini	Return only "number" or "unknown": How long has {name} has been claiming welfare benefits. Return the number of years as an integer. Return "unknown" if there is not enough information to determine the number of years. Do not guess.
Won the lottery	GPT-4o-mini	Return only "yes" or "no": According to this article, did {name} win the lottery? Return "no" if it is not mentioned.

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Classifier	Model	Prompt
Race (UK; input is the name only)	o4-mini	You are assisting academic researchers studying racial bias in UK media coverage. Given a person's name, predict the most likely broad racial category as perceived by a reader: respond with only "white" or "non-white". If genuinely ambiguous, respond "unclear". People of Roma, Romanian, European, or Eastern European origin should be classified as "white". People of Middle Eastern origin should be classified as "non-white". This is analogous to established name-based ethnicity classification methods used in public health and social science research (e.g. ONS Onomap, EthnicityEstimator).
<i>Sun sample filters</i>		
Reader communication	GPT-4o-mini	Respond only 'yes' or 'no': is this news article a communication (eg. letter, email, sms, tweet) from a reader of the newspaper to the newspaper? Only respond 'yes' if it explicitly says so, and respond 'no' otherwise. If it is about a communication, but that communication was not sent to the newspaper, respond 'no'.
Immigrants	GPT-4o-mini	Respond only 'yes' or 'no': In this article, identify the main individual or group claiming welfare benefits, receiving government support, or being discussed in relation to benefits. Is this individual or group described as having moved from another country to live in the UK or Ireland (i.e. an immigrant)? Answer 'yes' only if the text clearly states that they immigrated, are seeking asylum, were born abroad, came from another country, or have a foreign nationality. If the article does not mention immigration or country of origin, or if you are not sure, answer 'no'. Do not infer immigration status from the person's name, job, income, or stereotypes. This article is from a UK or Irish newspaper.
Jihadists or terrorists	GPT-4o-mini	Respond only 'yes' or 'no': is this news article mainly about jihadists or terrorists?
Related to the UK	GPT-4o-mini	Respond only 'yes' or 'no': is this news article related to the UK? This article is from a UK or Irish newspaper. If it is not clear, respond 'yes'. If it is about the Republic of Ireland, respond 'no'. If it is about Northern Ireland, respond 'yes'.

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Classifier	Model	Prompt
TV program	GPT-4o-mini	Respond only 'yes' or 'no': is this news article mainly about a TV programme, or individuals from a TV programme?
<i>Tropes</i>		
Article summary	o4-mini	Summarise this article in English in 25 words or fewer, without using any names of people, places or organisations. Do not mention what type of article it is.
Trope classifier (header of the system prompt, followed by the trope definitions of table S1)	GPT-5.4	You are a text classifier. Given a news article summary, classify it into exactly one of the following 14 clusters (c1-c14) or assign c0 if none fit. Focus on the core narrative of the article, not incidental details. A passing mention of a feature (e.g. a lenient sentence, a government job title, an immigrant) is not enough to assign the corresponding cluster - that feature must be central to the story. If the core narrative does not match any cluster, assign c0 even if incidental details could match a cluster. If an article fits more than one cluster, assign whichever cluster captures the primary focus of the narrative.

S3 Labelling rules

This section gives the labelling rules for the two finetuned classifiers described in appendix B.

S3.1 ‘Mentions benefits’ classifier

Any mention of working-age welfare benefits or the welfare system was considered on topic. In general, we considered a working-age welfare benefit to be any money or vouchers paid regularly by the state to an individual or family. This definition includes:

- Unemployment benefits
- Disability benefits
- Housing benefits
- Income assistance

- Income-based tax credits (e.g. Working Tax Credit, Child Tax Credit in the UK; Earned Income Tax Credit in the US)
- General discussions of the welfare state

The following do not count as welfare benefits under our definition:

- Pensions and other benefits for the elderly are not included. In the US and France, we exclude social security in general, unless the article specifies that it is not talking about retirement benefits.
- Student loans or grants, including Education Maintenance Allowance (UK).
- In-kind benefits, such as social housing, are not included, whereas housing benefits are.
- COVID-19 stimulus checks and other one-off payments are not included. Neither are furlough payments in the UK, as these were paid to employers, not individuals.
- Healthcare benefits or insurance (e.g. Medicare/Medicaid/ACA) are not included.

S3.2 ‘Mainly about benefits’ classifier

“Mainly about” is inherently subjective, so to ensure consistency across labellers, we provided them with the following rules of thumb:

- If it is only about benefits, then it is “mainly about”.
- If it mentions benefits and the only reason to write it is some message about benefits, then it is “mainly about”.
- If the article covers multiple different stories/statements, then at least 50% of them need to be on topic for it to be “mainly about”.
- If a very short summary would mention benefits it should be “mainly about”.
- If the headline mentions benefits then it is very likely that it is “mainly about”.